
Final Report

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1 Introduction

Eye diseases such as choroidal neovascularization (CNV), diabetic macular edema (DME), and drusen are among the leading causes of vision loss worldwide [1], [2]. Many of these diseases affect the retina and can be detected using medical imaging techniques such as Optical Coherence Tomography (OCT). OCT scans produce high-resolution cross-sectional images of the retina that allow doctors to observe structural changes in eye tissue [3]. However, analyzing these scans requires specialized expertise and can be time-consuming when large numbers of images must be examined [4]. Because of this, automated classification systems that can assist with identifying retinal diseases from OCT images could help reduce the workload for clinicians and support earlier detection of vision-threatening conditions.

The goal of this project is to develop a deep learning model that can automatically classify OCT images into four categories: Normal, CNV, DME, and Drusen. The input to the model is a 224×224 OCT image (Appendix A), and the output is the predicted disease class associated with that image. The task can therefore be formulated as a multi-class image classification problem where the model maps an OCT image to one of four diagnostic labels. Figure 1 shows the overall workflow of the system. This problem is well suited for deep learning because it is fundamentally an image classification task. Convolutional Neural Networks (CNNs) are designed to learn visual patterns directly from images by identifying features such as edges, textures, and structural shapes. In OCT scans, diseases often appear as subtle structural changes in the retinal layers, which makes CNNs particularly effective at detecting these patterns automatically [3]. In this project, deep learning models will be used to learn these visual features from OCT images and classify different types of retinal disease.

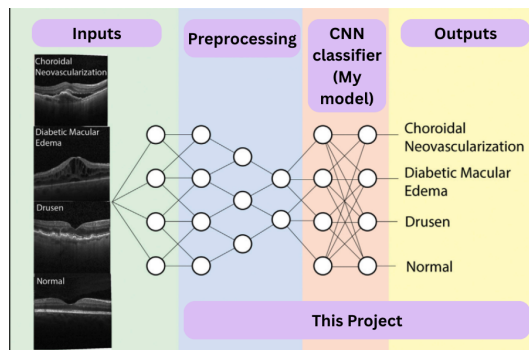


Figure 1: Overview of the retinal OCT disease classification pipeline. Preprocessed OCT images are input into a deep learning model that extracts hierarchical visual features and classifies each image into one of four categories: CNV, DME, Drusen, or Normal.

2 Background & Related Work

Convolutional Neural Networks (CNNs) form the foundation of modern image classification systems. LeCun et al. [5] introduced early convolutional networks, establishing key concepts such as weight sharing and local receptive fields. Krizhevsky et al. [6] later demonstrated that deep CNNs significantly outperform traditional methods, achieving breakthrough on ImageNet. Subsequent work by Simonyan and Zisserman [7] and He et al. [8] further improved performance through deeper architectures (VGGNet) and residual connections (ResNet), enabling more effective feature

learning. Deep learning has also shown strong performance in medical imaging. Gulshan et al. [9] demonstrated that CNNs can match expert-level performance in detecting retinal diseases from medical images. Most directly related to this work, Kermany et al. [3] applied transfer learning to classify OCT images into four retinal disease categories, achieving over 96% accuracy and releasing the dataset used in this project. Their work highlights the effectiveness of pretrained CNNs for OCT-based diagnosis. This project builds on these prior works by applying transfer learning with AlexNet to the OCT classification task. Compared to deeper architectures, AlexNet provides a simpler and computationally efficient model while still leveraging pretrained visual features. The model is compared against a baseline MLP to demonstrate the importance of spatial feature extraction and transfer learning for medical image classification.

3 Data Processing

The dataset used in this project is derived from the publicly available retinal OCT collection introduced by Kermany et al. [3]. A smaller subset of 4,289 images obtained from a Kaggle repository [10] was used for training, validation, and testing. This subset provides a balance between computational efficiency and sufficient data for model development. The dataset contains four classes: CNV, DME, Drusen, and Normal. It was split into 70% training, 15% validation, and 15% test sets using stratified sampling (random state=22) to preserve class proportions across all splits (Appendix B). The test set was held out during training and hyperparameter tuning. Table 1 shows the class distribution across each split. All images were loaded using PIL and converted to grayscale. To match the input requirements of AlexNet, grayscale images were replicated to three channels and resized to 224×224 pixels. Pixel values were normalized to the range $[-1, 1]$. Data augmentation was applied to training images using random horizontal flipping ($p=0.5$) and random rotation (± 10) to improve generalization. Validation and test images used only resizing and normalization. OCT images exhibit distinct structural patterns across classes, such as fluid accumulation in CNV and localized deposits in Drusen, making the task well suited for convolutional models that learn spatial features [3].

Class	Train	Validation	Test	Total
CNV	871	187	187	1245
DME	622	133	133	888
Drusen	745	159	160	1064
Normal	764	164	164	1092
Total	3002	643	644	4289

Table 1: Stratified class distribution across all splits (total: 4289 images)

4 Architecture

The final model used in this project is based on AlexNet pretrained on ImageNet and fine-tuned for retinal OCT classification. The network takes as input an image of size $3 \times 224 \times 224$. The convolutional layers of AlexNet, which consist of stacked convolution, ReLU activation, and pooling operations, were used as a fixed feature extractor. These layers were frozen during training to preserve pretrained visual features such as edges, textures, and spatial patterns. The original classifier head was modified by replacing the final fully connected layer with a new linear layer of size *Linear(4096, 4)*, corresponding to the four output classes (Normal, CNV, DME, and Drusen). Only the classifier layers were trained, while all convolutional layers remained fixed. The model was trained using cross-entropy loss and the Adam optimizer with a learning rate of 0.001 and weight decay of 0.0001, applied only to the classifier parameters. Training was performed for up to 20 epochs, with early stopping (patience=5) based on validation performance. This architecture is well suited for OCT image classification because convolutional layers preserve spatial structure and learn hierarchical features, enabling the model to detect subtle retinal abnormalities. By leveraging pretrained features and limiting the number of trainable parameters, the model achieves strong performance while reducing overfitting on a relatively small dataset.

5 Baseline Model

A simple multilayer perceptron (MLP) was implemented as a baseline to establish a lower bound on performance. Each $3 \times 224 \times 224$ input image was flattened into a vector of length 150,528 and passed through a fully connected network consisting of one hidden layer with 128 units (ReLU activation) and Dropout ($p=0.3$), followed by a 4-unit output layer corresponding to the four classes. The model was trained using cross-entropy loss and the Adam optimizer ($\text{lr}=0.001$) for up to 20 epochs. The baseline achieved approximately 29% validation accuracy and largely predicted a single dominant class, indicating that it failed to learn meaningful distinctions between classes. Compared to earlier experiments using lower-resolution inputs (128×128), performance decreased due to the increased input dimensionality in the final setup, highlighting the limitations of fully connected networks for high-dimensional image classification tasks.

6 Quantitative Results

Table 2 compares the baseline MLP and AlexNet transfer learning models. AlexNet achieves 83.8% validation accuracy and 88.5% test accuracy, compared to 29.4% for the MLP (59-point improvement). The test accuracy exceeds validation accuracy, indicating good generalization and that the stratified split was representative. Per-class performance on the test set (Table 3) shows that CNV achieves the highest F1-score (0.94), consistent with its distinctive structural features. DME achieves $F1=0.83$ and Drusen $F1=0.86$, reflecting more subtle visual differences, while the Normal class reaches $F1=0.88$. Appendix C shows the training dynamics. Validation accuracy slightly exceeds training accuracy in early epochs, which is characteristic of transfer learning where pretrained features generalize effectively from the start. Both curves converge around epoch 10, and early stopping is triggered at epoch 15, restoring the best-performing model. In contrast, the baseline model failed to learn meaningful class distinctions and frequently predicted a single dominant class (Appendix D).

Table 2: Accuracy comparison between baseline MLP and AlexNet transfer learning models.

Model	Validation Accuracy	Test Accuracy	Parameters
Baseline MLP	29.4%	29.0%	19.3M
AlexNet Transfer	83.8%	88.5%	57.0M

Table 3: Per-class precision, recall, and F1-score for the AlexNet model on the test set ($n = 644$).

Class	Precision	Recall	F1-score	Support
CNV	0.91	0.97	0.94	187
DME	0.93	0.76	0.83	133
Drusen	0.90	0.82	0.86	160
Normal	0.82	0.95	0.88	164
Macro Avg	0.89	0.88	0.88	644
Weighted Avg	0.89	0.89	0.88	644

7 Qualitative Results

Appendix E shows eight representative predictions from the test set, selected to illustrate both correct classifications and common failure cases. The model correctly identifies CNV scans exhibiting subretinal fluid and irregular structures beneath the retinal pigment epithelium, as well as Normal scans characterized by smooth and well-defined retinal layers. The most common errors occur between DME and Normal, consistent with the confusion matrix (Appendix G-H). Mild DME cases often contain only subtle intraretinal fluid, which can visually resemble a normal retina at 224×224 resolution. Similarly, confusion between Drusen and Normal is observed, as early drusen deposits produce only small, localized irregularities in the retinal layers that are difficult to distinguish. These qualitative results support the quantitative findings that the model performs well on classes with

distinct structural features (CNV), while performance is lower on classes with subtle or overlapping visual patterns (DME, Drusen).

8 Evaluation on New Data

Although an independent external dataset was not used, model performance was evaluated using a held-out test set that was not used during training or hyperparameter tuning. The dataset was split using a 70/15/15 stratified split, ensuring that class distributions were preserved across training, validation, and test sets. The close agreement between validation accuracy (83.8%) and test accuracy (88.5%) indicates good generalization and suggests that the model does not overfit to the training data. Although evaluation on external dataset would provide a stronger assessment, this setup provides a reliable estimate of performance on unseen OCT images.

9 Discussion

The results demonstrate that transfer learning with AlexNet is highly effective for OCT image classification, achieving 88.5% test accuracy, a substantial improvement over the baseline MLP (29%). This confirms that preserving spatial structure through convolutional layers is essential for this task, as retinal diseases are defined by localized anatomical patterns that cannot be captured by fully connected networks. One notable observation is the large performance gap between the baseline and the CNN, despite the MLP having a comparable number of parameters. This shows that model architecture is more important than parameter count for image-based tasks. Additionally, the decrease in baseline performance compared to previous experiment (48.67% at 128×128 resolution) further emphasizes that fully connected models scale poorly with increasing input dimensionality. The model performs best on the CNV class, which aligns with its high F1-score (0.94) and reflects the presence of distinct structural abnormalities such as subretinal fluid. In contrast, lower performance on DME and Drusen suggests that these conditions are more difficult to distinguish due to their subtle and overlapping visual features. This is also supported by the confusion matrix and qualitative examples, where misclassifications occur primarily between these classes. An interesting observation is that validation accuracy is slightly lower than test accuracy. While this may seem counterintuitive, it suggests that the validation set may be slightly more challenging or that the stratified split produced a representative test set. The stable training curves and lack of overfitting further support that the model generalizes well. Despite strong performance, the model has several limitations. The dataset size is relatively small, and all data originates from the same source, which may limit generalization to real-world clinical settings. Additionally, resizing images to 224×224 may remove fine-grained details that are important for distinguishing subtle conditions such as early-stage DME or Drusen. Overall, this project addresses a challenging medical image classification task involving subtle and overlapping retinal features. Despite the limited dataset size, the model achieves 88.5% test accuracy, demonstrating that transfer learning effectively captures meaningful spatial patterns. Future work could explore larger datasets, fine-tuning deeper layers, or using more advanced architectures to further improve its performance.

10 Ethical Considerations

The use of deep learning for medical image classification raises important ethical concerns. Although the model achieves strong performance, incorrect predictions, particularly false negatives, could delay diagnosis and treatment, potentially leading to serious consequences for patients. As a result, this system should be used as a decision-support tool rather than a replacement for clinical judgment. There are also limitations related to the training data. The dataset is derived from a single source and may not represent the full diversity of patient populations, imaging devices, or clinical conditions. This could introduce bias and reduce the model’s reliability when applied in different settings. Additionally, subtle conditions such as early-stage DME are more difficult for the model to detect, which may disproportionately affect certain patient groups. In conclusion, while the model shows promising performance, careful validation on diverse clinical data and integration with expert oversight are necessary to ensure safe and equitable use.

References

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Appendix

A Sample OCT Images

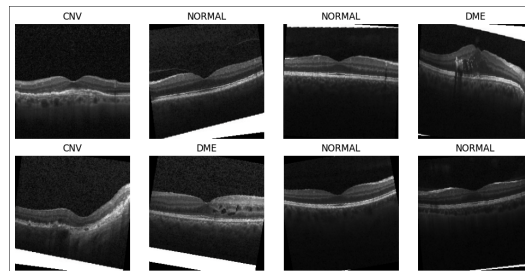


Figure 2: Sample grayscale OCT images from the dataset.

B Class Distribution

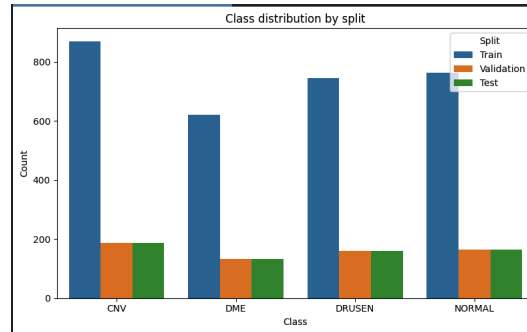


Figure 3: Class distribution by split.

C Training Curves (Transfer Learning)

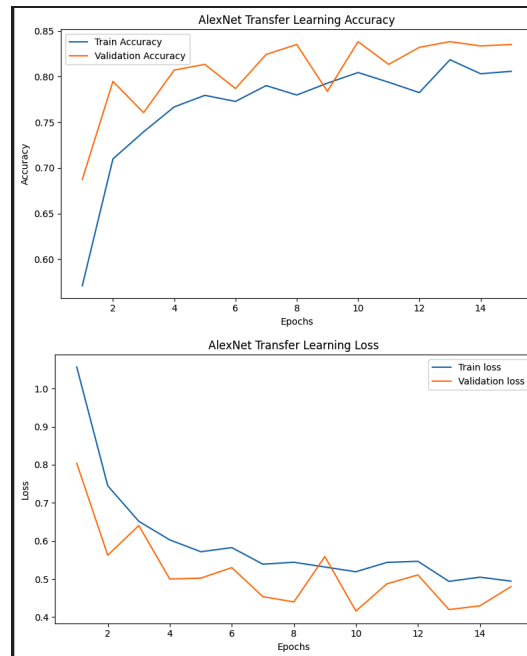


Figure 4: AlexNet training and validation accuracy (top) and loss (bottom) over 15 epochs. Validation accuracy leads training accuracy in early epochs, characteristic of transfer learning. Early stopping restored the best checkpoint.

D Training Curves (Baseline)

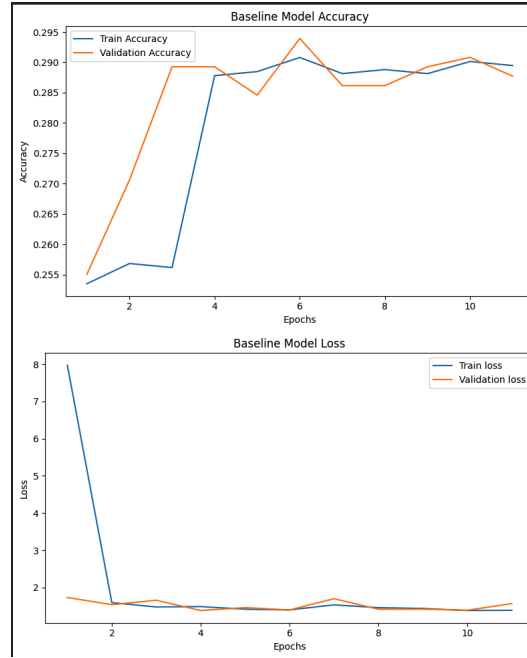


Figure 5: Training and validation accuracy (top) and loss (bottom) for the baseline MLP model. The model quickly plateaus and shows limited improvement, indicating poor learning of spatial features.

E Sample Predictions

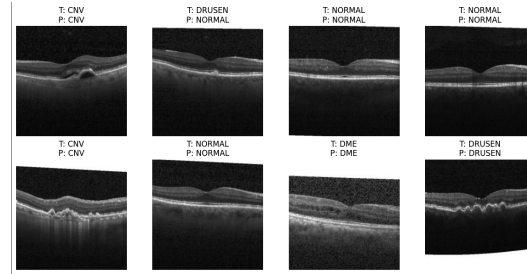


Figure 6: Representative predictions from the test set (T=true label, P=predicted label). The model reliably identifies CNV scans (with visible subretinal fluid) and Normal scans. Errors concentrate in DME cases that resemble Normal, consistent with the confusion matrix.

F Baseline Confusion Matrix

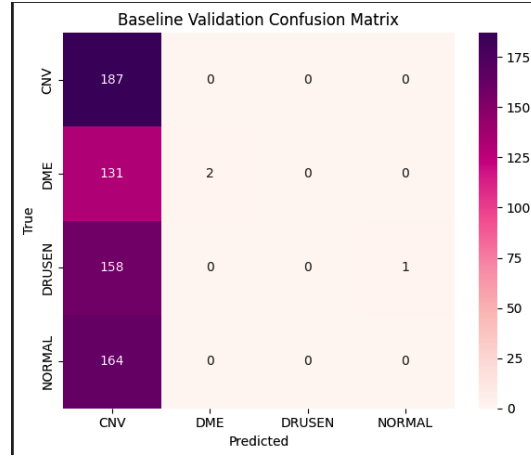


Figure 7: Confusion matrix for the baseline MLP model on the validation set. The model predicts the majority of samples as CNV, indicating a failure to learn meaningful class distinctions and severe class imbalance bias.

G Transfer Learning Confusion Matrix (Validation)

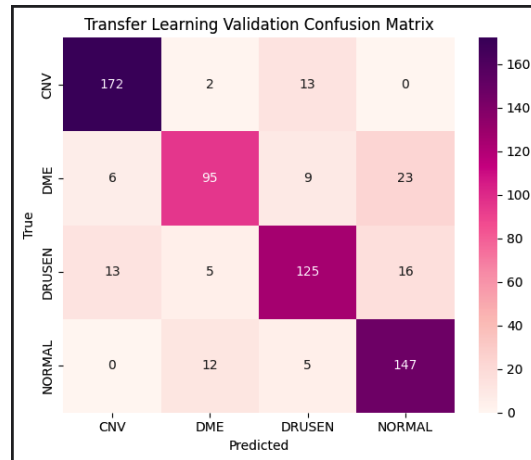


Figure 8: Confusion matrix for the AlexNet transfer learning model on the validation set. The model demonstrates balanced performance across classes, with minor confusion between DME and Normal due to subtle structural similarities.

H Transfer Learning Confusion Matrix (Test)

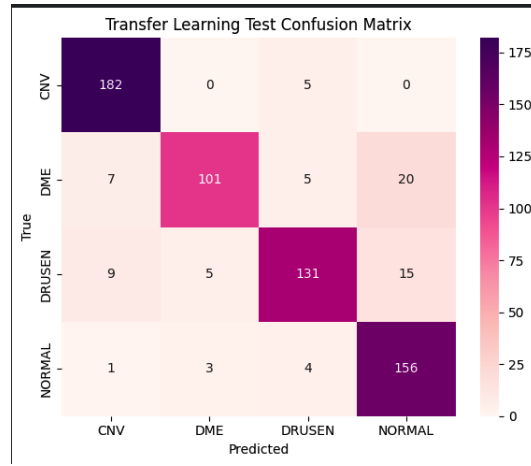


Figure 9: Confusion matrix for the transfer learning model on the test set. The model achieves strong performance across all classes, with most predictions concentrated along the diagonal. Misclassifications primarily occur between DME and Normal, reflecting their visual similarity.